



Natural Language Processing Project Proposal

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Course **Natural Language Processing** Section _____ Group number **5**

Name	Student ID	Main responsibilities
Sameer Almadani	2230001818	Data cleaning and split, Method 1 (TF-IDF), GitHub
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Mohaiya Aldossary	2230004579	Negation test, error analysis, report and slides

Selected Option and Purpose

Option 2: Review Sentiment Analysis. Hotels get many online reviews every day and staff cannot read all of them. Our notebook reads a hotel review and says if the guest was happy or not, so a hotel manager or customer service team can find unhappy guests fast and reply to them.

Input and Output

Input: one English hotel review (a sentence or a short paragraph). **Output:** Positive or Negative, with a confidence score. Example: “The room was clean and the staff were friendly” → Positive. “The room was not clean and nobody helped us” → Negative. Mixed or neutral reviews do not fit two labels, and we will discuss this as a limitation.

Dataset and Size

Datafiniti Hotel Reviews [1], a public dataset on Kaggle with 10,000 reviews of US hotels and a 1–5 star rating. We label 1–2 stars as Negative and 4–5 stars as Positive and drop 3-star reviews. After removing empty and duplicate reviews we have about 8,400 reviews (about 1,150 Negative and 7,270 Positive). We split them 70% train, 15% development and 15% test, stratified, with duplicates removed first so the same review cannot be in two splits. The data has many more positive reviews, so macro F1 is our main score.

Two Methods

- **Method 1, traditional:** TF-IDF word features (single words and word pairs) with logistic regression, trained on our training set. Word pairs like “not clean” give it some chance to catch negation. We pick its settings on the development set.
- **Method 2, pretrained:** the RoBERTa sentiment model cardiffnlp/twitter-roberta-base-sentiment [2], used as it is with no extra training. We turn its output into one positive score and pick the cut-off on the development set.

Proposed Challenge and Experimental Question

Challenge: negation. Guests write things like “not worth the price”, “no hot water” or “wasn’t bad”, where the meaning is the opposite of the sentiment word. **Question:** Which method gets a higher macro F1 on hotel reviews that contain negation words (no, not, never, neither, nor, without, cannot, n’t): TF-IDF with logistic regression trained on hotel reviews, or the pretrained RoBERTa model?

We fix the negation word list before testing. Both methods are tested on the same test reviews with accuracy and macro F1, on the full test set and on reviews with and without negation, and compared with a majority-class baseline. We will explain at least three wrong predictions, and the Try it cell will classify any new review without retraining.

Related Work

Pang et al. [3] used word features with naive Bayes, maximum entropy (logistic regression) and SVM classifiers on movie reviews and got strong results. They also tagged words that come after a negation. This is why we use TF-IDF with logistic regression and add word pairs. Wiegand et al. [4] reviewed many studies on negation in sentiment analysis and found that knowing which words a negation affects is hard, mostly for simple word-based models. The RoBERTa model we use comes from Barbieri et al. [2] and was trained on tweets, so we also check whether it works on longer hotel reviews as well as a model trained on hotel data.

References

- [1] Datafiniti, “Hotel Reviews,” Kaggle. <https://www.kaggle.com/datasets/datafiniti/hotel-reviews>
- [2] F. Barbieri, J. Camacho-Collados, L. Espinosa-Anke and L. Neves, “TweetEval: Unified benchmark and comparative evaluation for tweet classification,” in Findings of EMNLP, 2020.
- [3] B. Pang, L. Lee and S. Vaithyanathan, “Thumbs up? Sentiment classification using machine learning techniques,” in Proc. EMNLP, 2002, pp. 79–86.
- [4] M. Wiegand, A. Balahur, B. Roth, D. Klakow and A. Montoyo, “A survey on the role of negation in sentiment analysis,” in Proc. Workshop on Negation and Speculation in NLP, 2010, pp. 60–68.